

Categorical Data Guided Lab

Robert T Kohn

9/2/25

Load Arthritis Package

```
require(vcd)

## Loading required package: vcd
## Warning: package 'vcd' was built under R version 4.4.3
## Loading required package: grid
##
## Attaching package: 'vcd'
## The following object is masked from 'package:mosaic':
##
##      mplot
data("Arthritis")
head(Arthritis)

##      ID Treatment  Sex Age Improved
## 1  57   Treated Male  27      Some
## 2  46   Treated Male  29      None
## 3  77   Treated Male  30      None
## 4  17   Treated Male  32   Marked
## 5  36   Treated Male  46   Marked
## 6  23   Treated Male  58   Marked
```

Planning the exploration

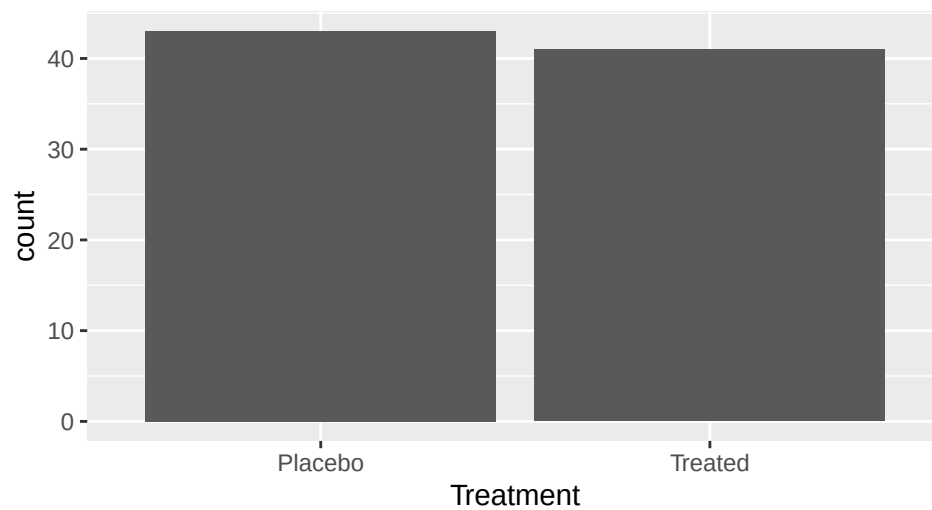
It is important to have a guided question for each visual because otherwise we can form biases off the data and form a hypothesis to match the work we have done which then causes the work to be flawed. Additionally, a hypothesis is fundamental with all statistical analysis.

I don't know how to embed an image into markdown but imagine a chart similar to yours but some is reduced for treatment.

Bar Charts

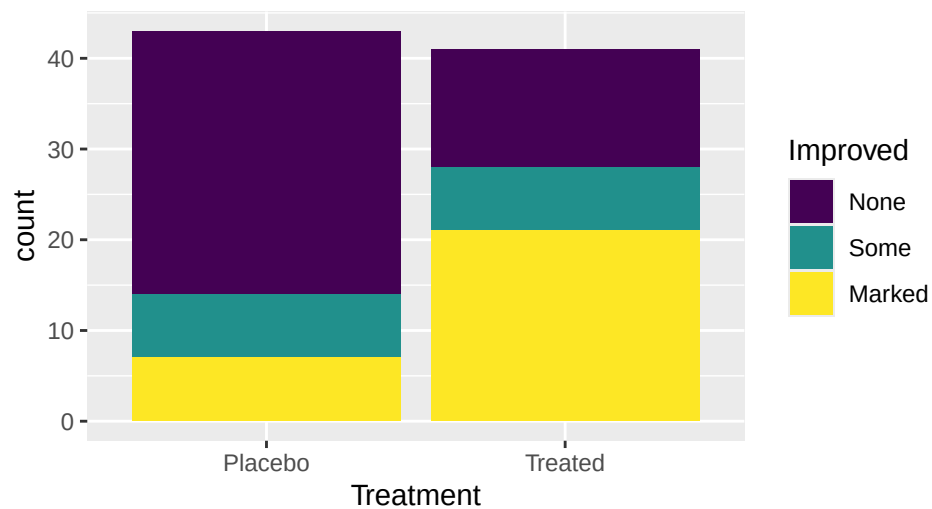
```
gf_bar( ~ Treatment , data = Arthritis, title="Arthritis Bar Chart")
```

Arthritis Bar Chart

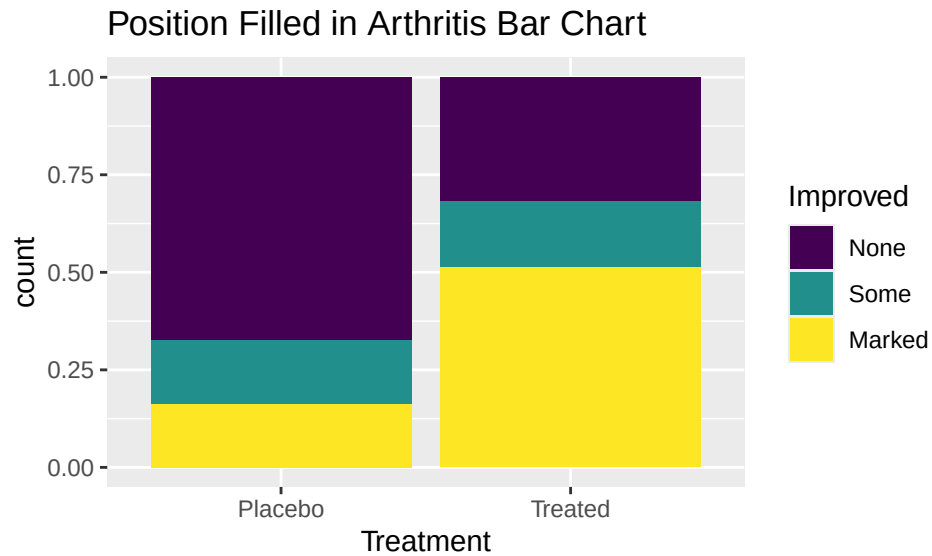


```
gf_bar( ~ Treatment , data = Arthritis , fill = ~ Improved, title="Filled in Arthritis Bar Chart" )
```

Filled in Arthritis Bar Chart

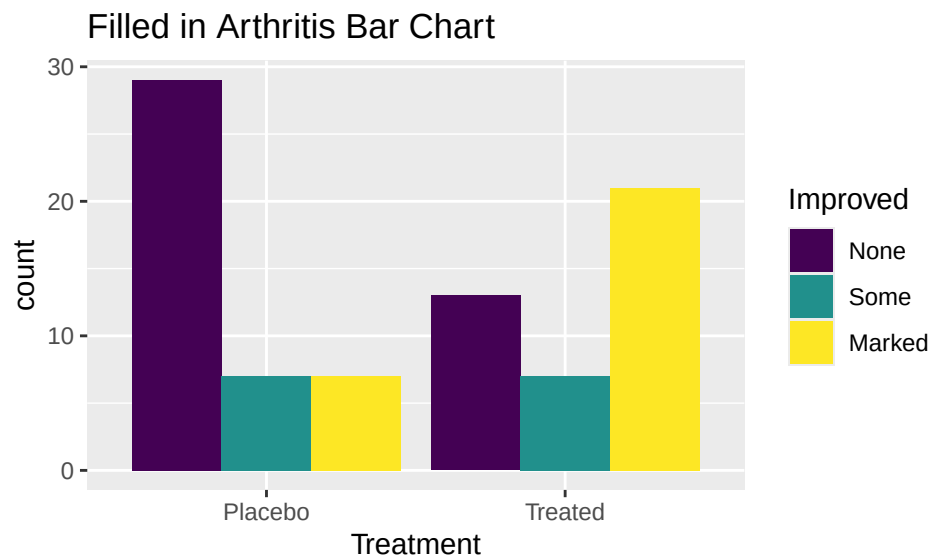


```
gf_bar( ~ Treatment , data = Arthritis , fill = ~ Improved , position = position_fill( ),title="Position" )
```

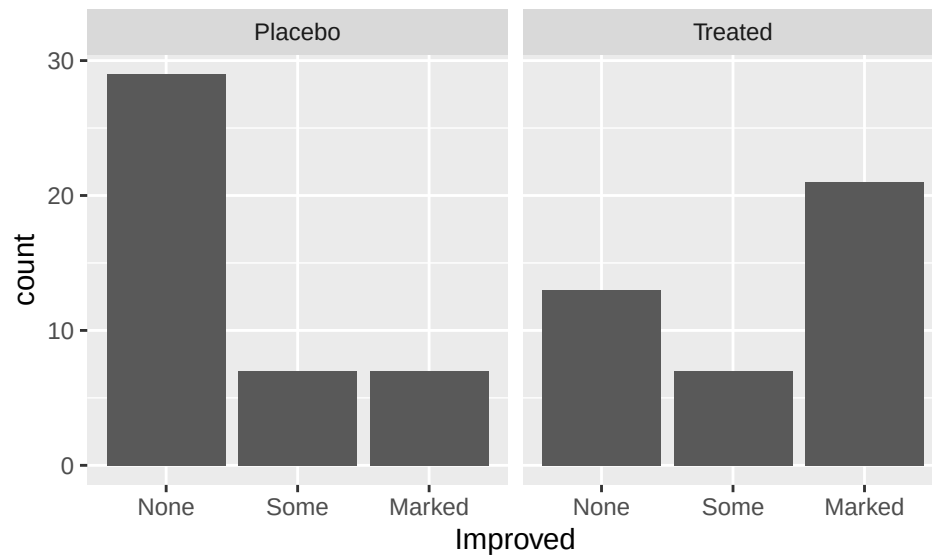


Side Bar Charts

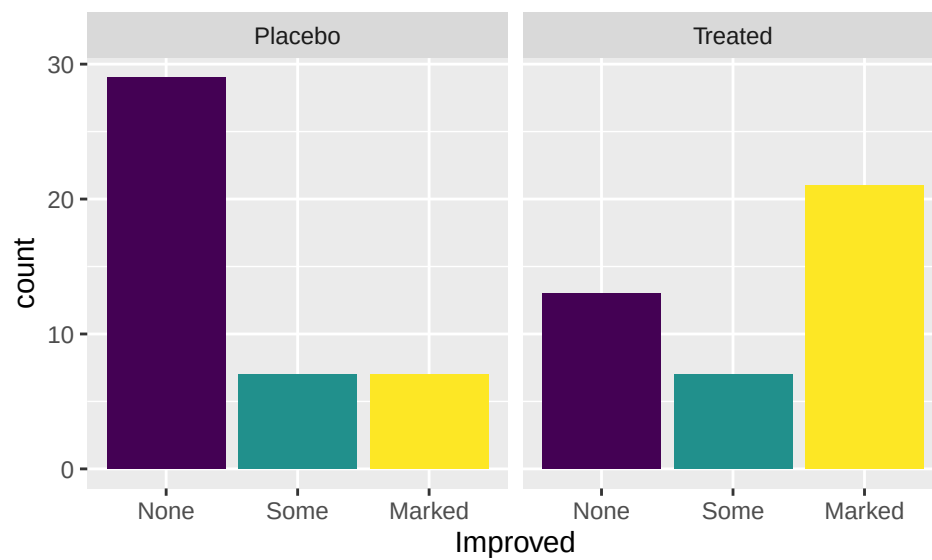
```
gf_bar( ~ Treatment , data = Arthritis , fill = ~ Improved, position = position_dodge( ), title="Filled in Arthritis Bar Chart")
```



```
gf_bar(~Improved, data = Arthritis) %>% gf_facet_wrap(~Treatment)
```

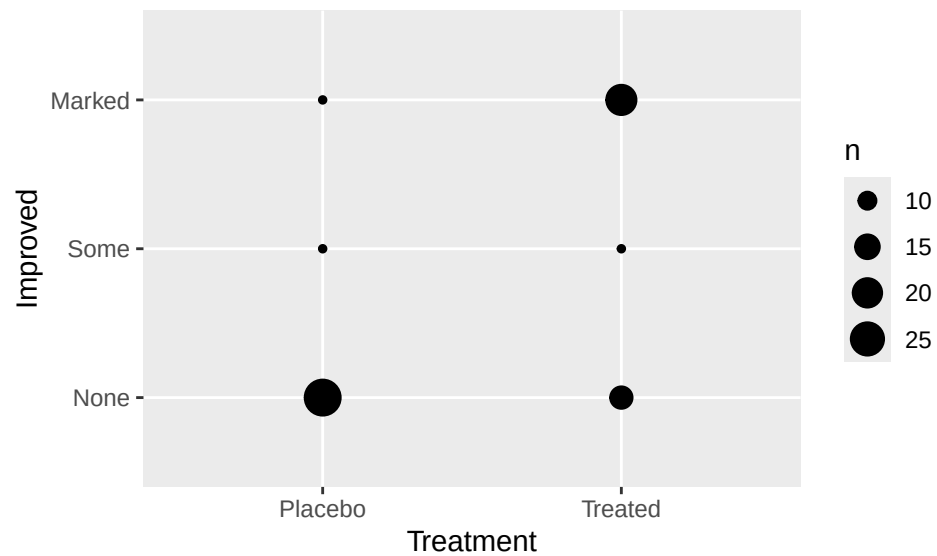


```
gf_bar(~Improved, data = Arthritis, fill = ~Improved, show.legend = FALSE) %>%
  gf_facet_wrap(~Treatment)
```

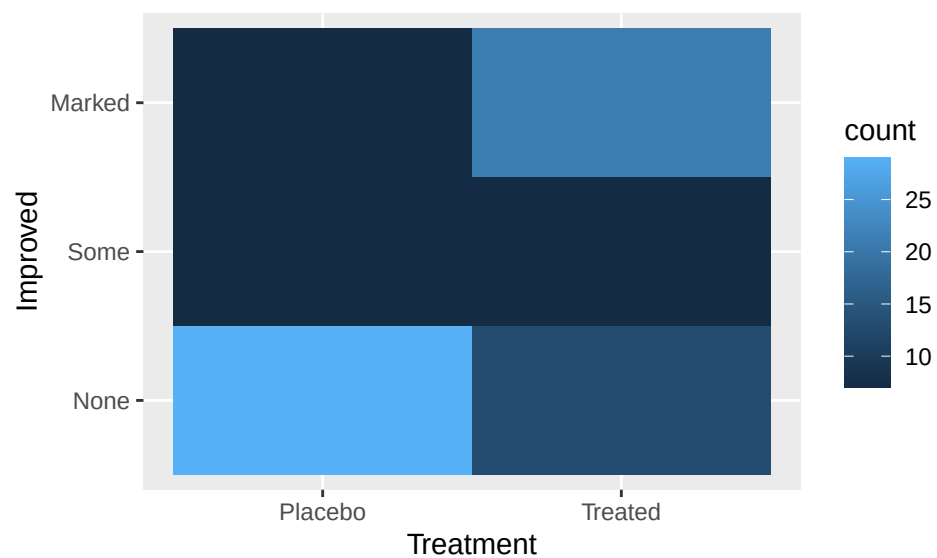


Alt Charts

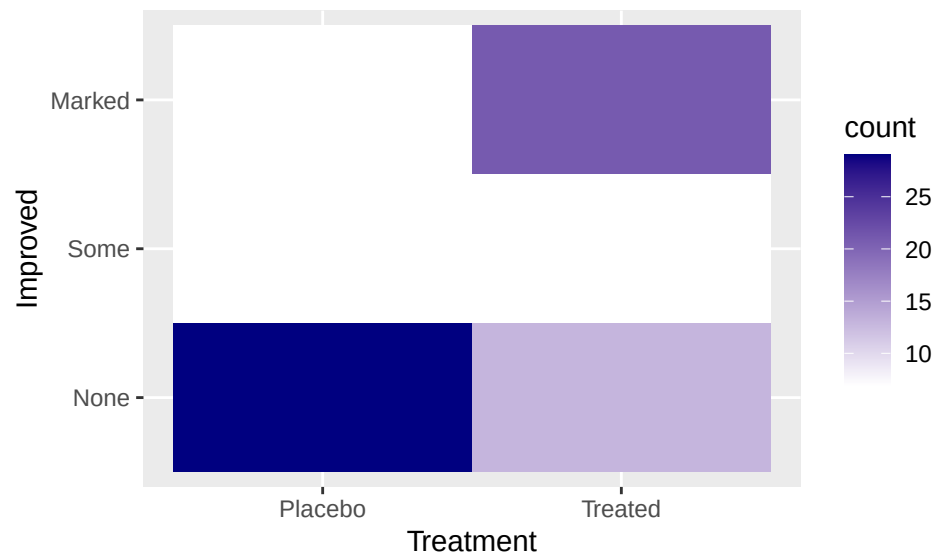
```
gf_count( Improved ~ Treatment, data = Arthritis )
```



```
gf_bin2d( Improved ~ Treatment , data = Arthritis )
```



```
gf_bin2d(Improved~Treatment, data=Arthritis) %>%
gf_refine(scale_fill_continuous(low = "white", high = "navy"))
```



Load Hair and Eye data

```
data("HairEyeColor")
HairEye <- as.data.frame(HairEyeColor)
head(HairEyeColor)
```

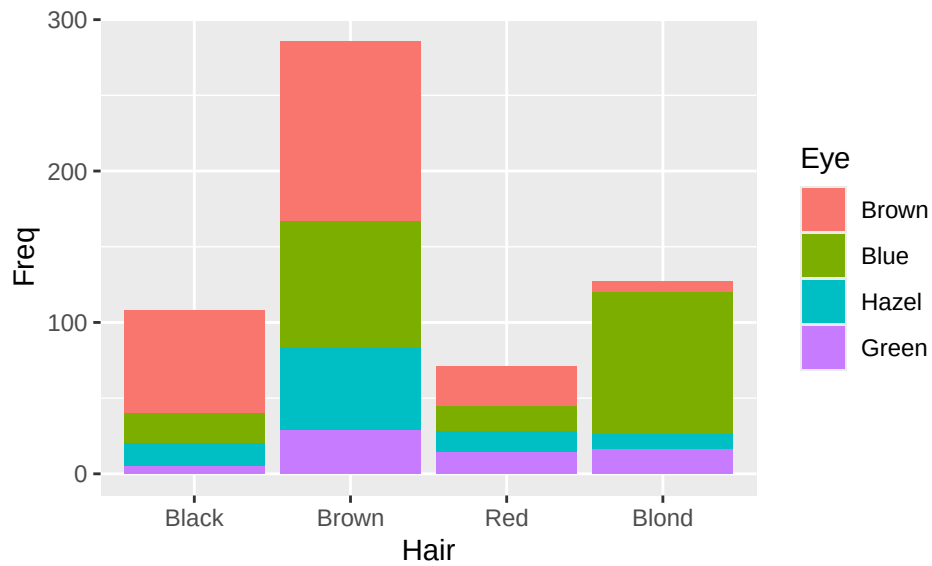
```
## , , Sex = Male
##
##      Eye
## Hair   Brown Blue Hazel Green
## Black   32   11   10    3
## Brown   53   50   25   15
## Red     10   10    7    7
## Blond    3   30    5    8
##
## , , Sex = Female
##
##      Eye
## Hair   Brown Blue Hazel Green
## Black   36    9    5    2
## Brown   66   34   29   14
## Red     16    7    7    7
## Blond    4   64    5    8
```

Planning Exploration

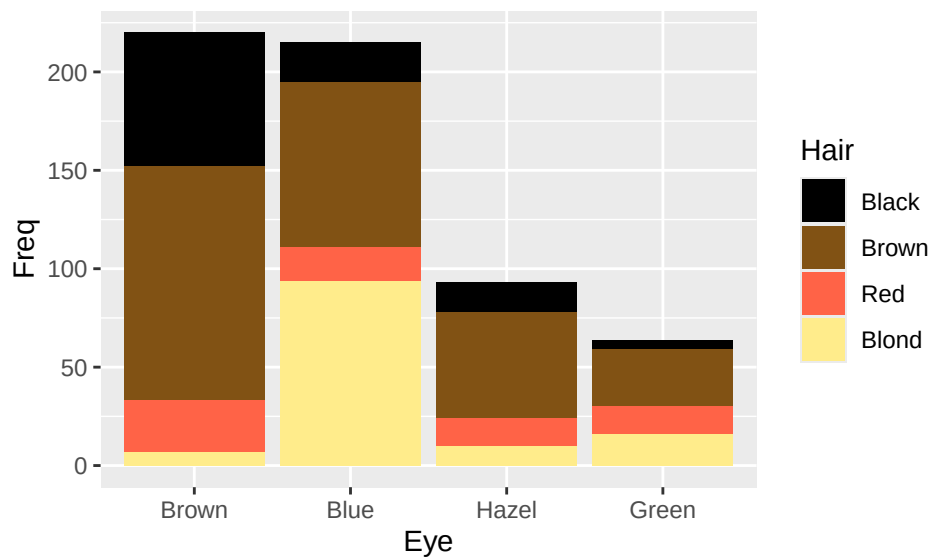
I have no clue how to embed yet again but I'm going to list each in descending order such that $\text{Hair} = \{\text{Eye}, \text{Eye}, \dots\}$. $\text{Black} = \{\text{Brown}, \text{Blue}, \text{Hazel}, \text{Green}\}$ $\text{Brown} = \{\text{Brown}, \text{Blue}, \text{Hazel}, \text{Green}\}$ $\text{Red} = \{\text{Brown}, \text{Blue}, \text{Hazel}, \text{Green}\}$ and $\text{Blond} = \{\text{Blue}, \text{Green}, \text{Hazel}, \text{Brown}\}$.

Column Chart

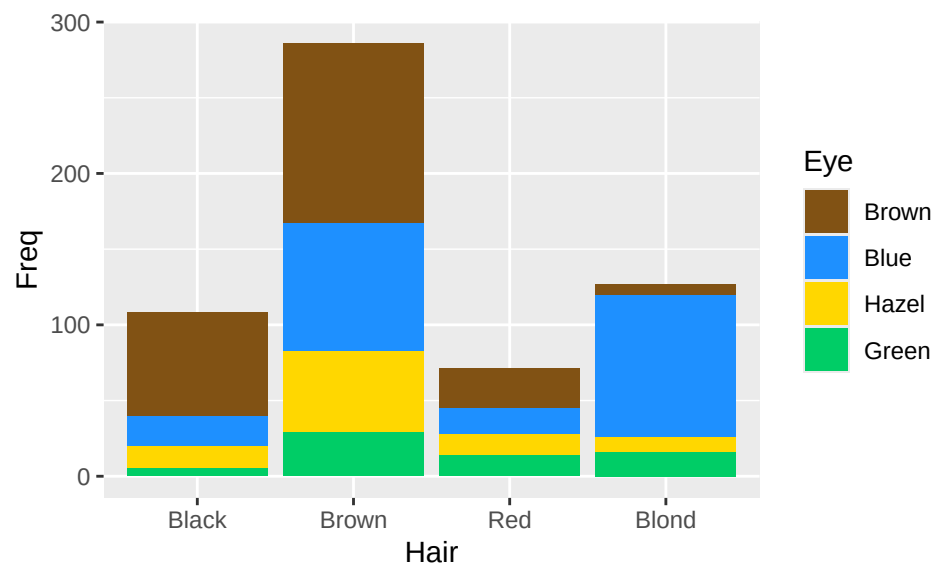
```
library(ggformula)
gf_col(Freq ~ Hair, data = HairEye, fill = ~Eye)
```



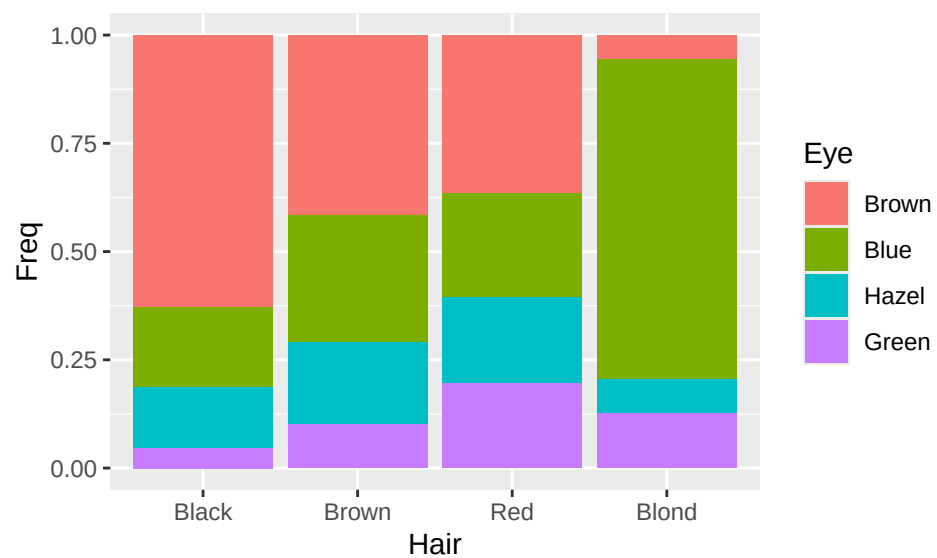
```
gf_col(Freq~Eye, data =HairEye, fill = ~Hair) %>%
gf_refine(scale_fill_manual(values = c("black", "#815214", "tomato", "lightgoldenrod1")))
```



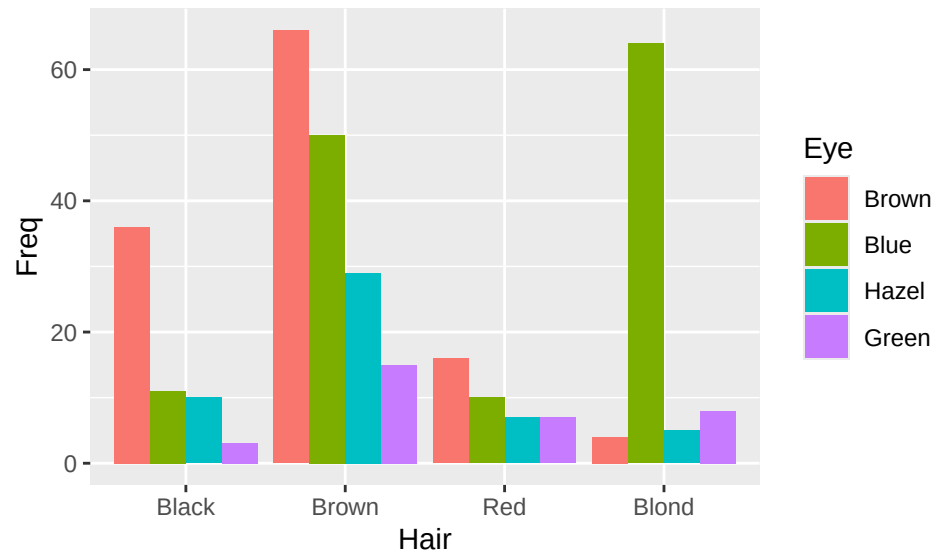
```
gf_col(Freq ~ Hair, data = HairEye, fill = ~Eye) %>% gf_refine(scale_fill_manual(values = c("#815214",
```



```
gf_col(Freq ~ Hair, data = HairEye, fill = ~Eye, position = position_fill())
```

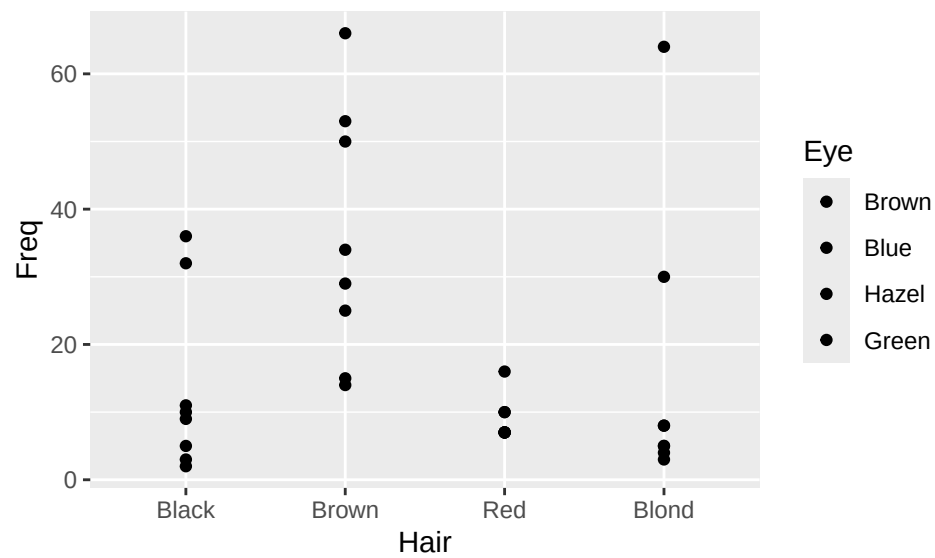


```
gf_col(Freq ~ Hair, data = HairEye, fill = ~Eye, position = position_dodge())
```

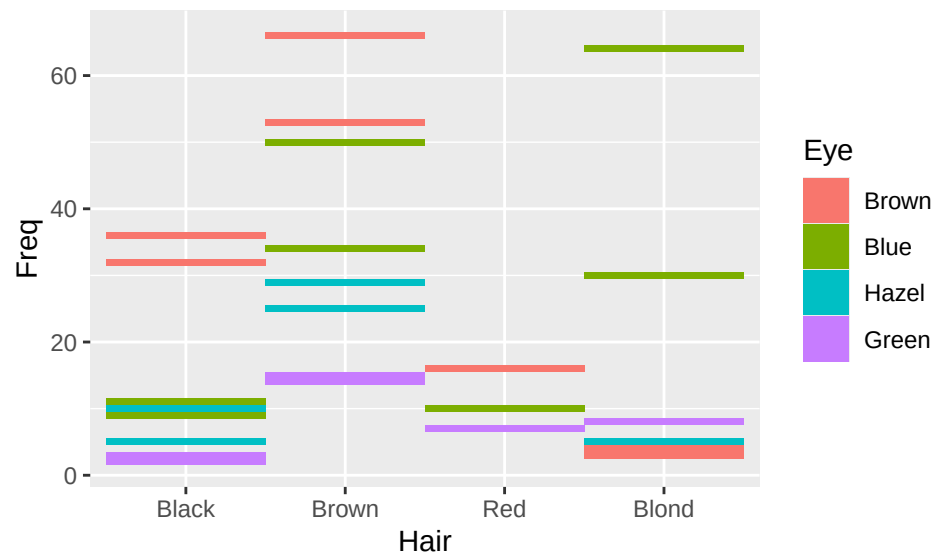


Count Chart

```
gf_point(Freq ~ Hair, data = HairEye, fill = ~Eye)
```



```
gf_tile(Freq ~ Hair, data = HairEye, fill = ~Eye)
```



```
gf_tile(Hair~Eye, fill=~Freq, data = HairEye) %>%
gf_refine(scale_fill_continuous(low = "white", high = "navy"))
```

